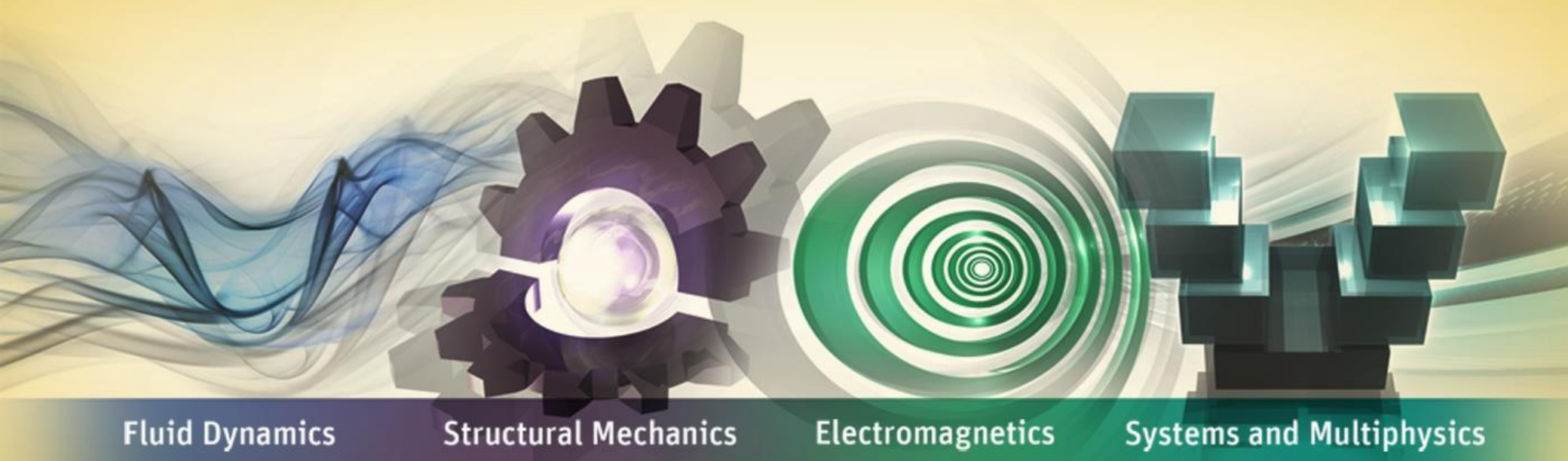


Workshop 5B: Post Buckling Analysis

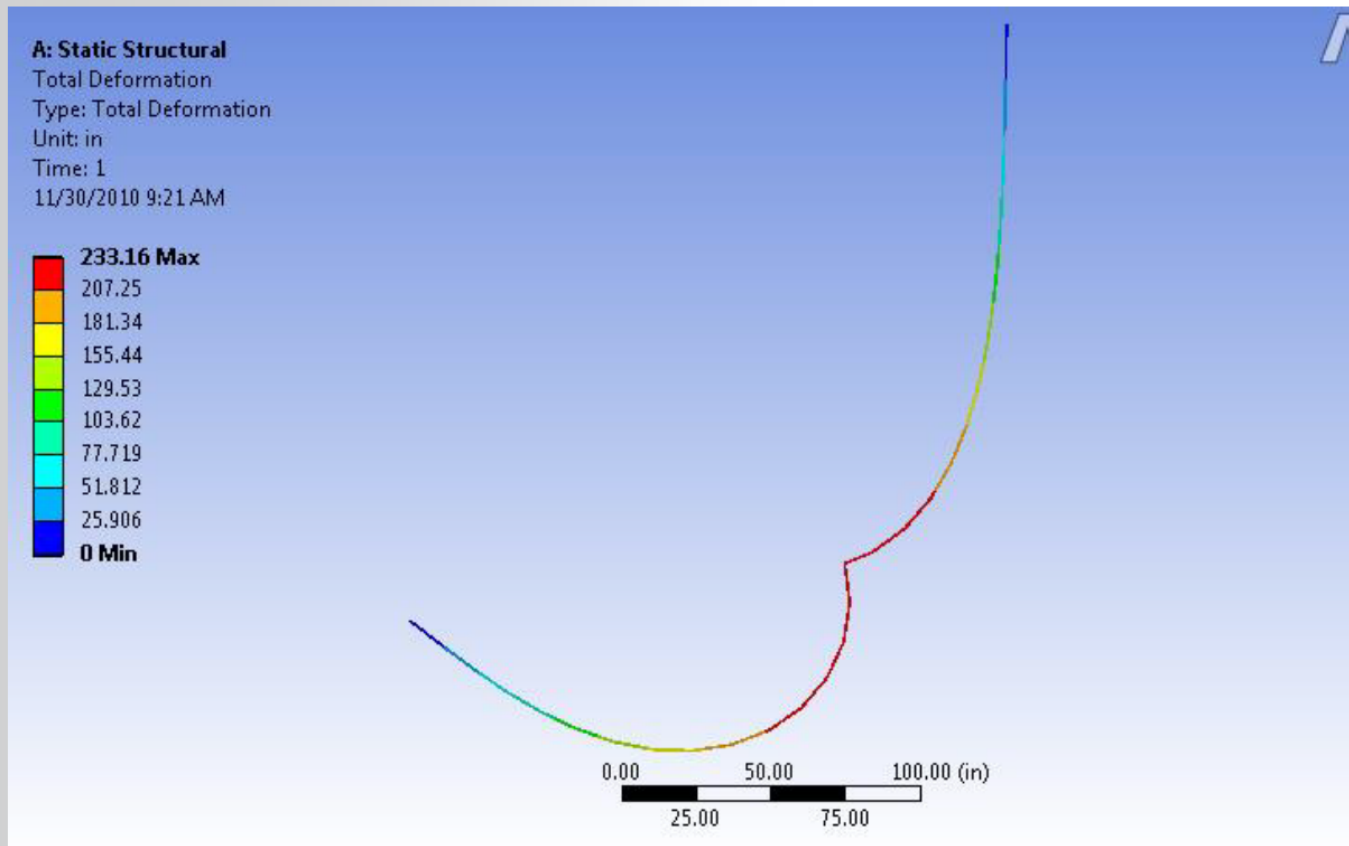


ANSYS Mechanical Introduction to Structural Nonlinearities

Workshop 5B: Post Buckling

Goal

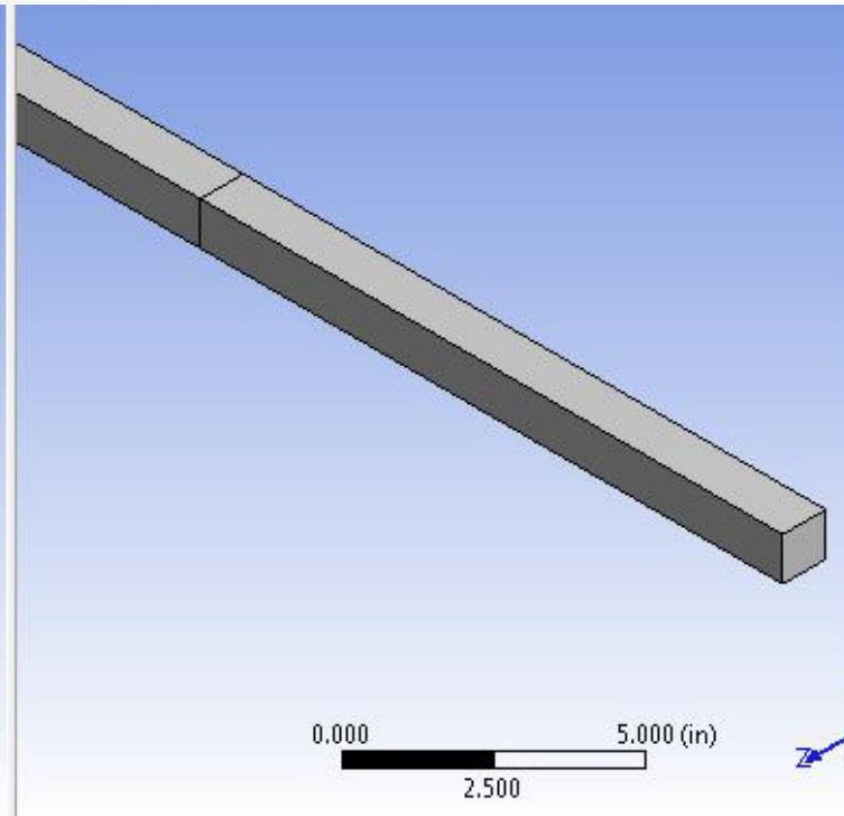
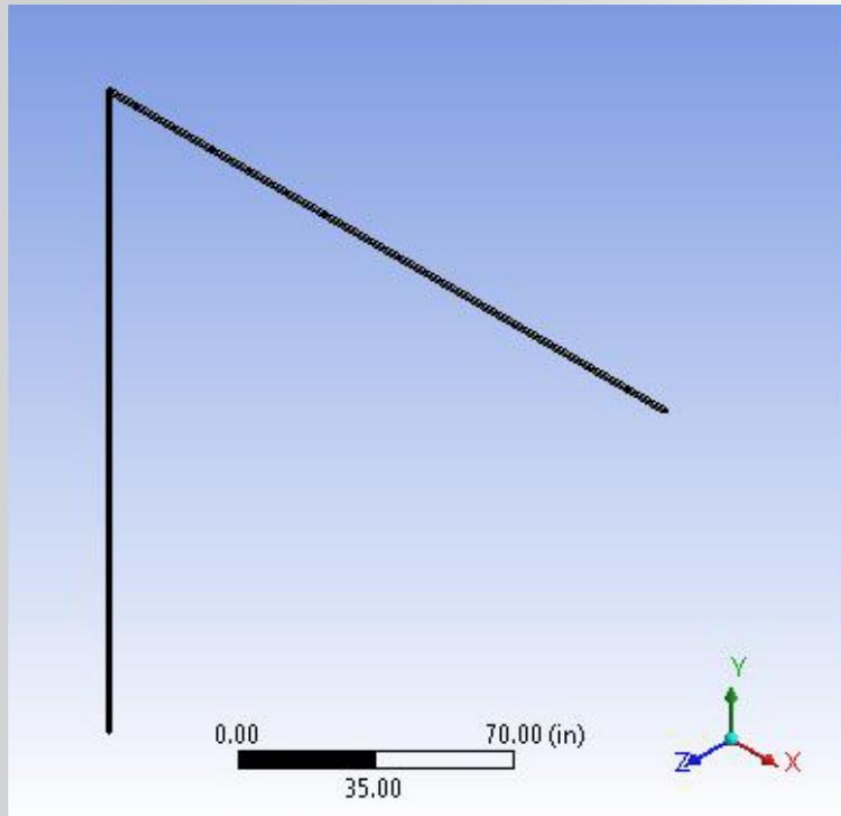
Use Nonlinear Stabilization tools together with restart technology to simulate post buckling behavior.



...Workshop 5B: Post Buckling

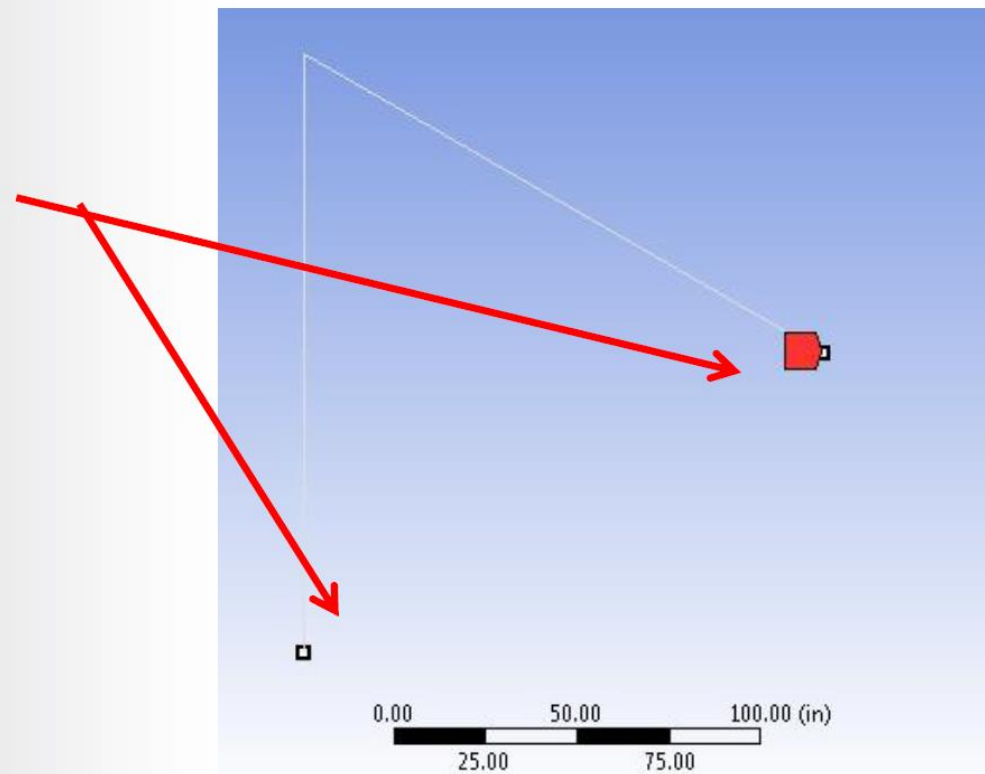
- Model Description (Same model as Workshop 5A)

- L shaped frame represented with 3D Line Bodies
- Meshed with 3D Beam elements
- Linear Elastic Material



- Model Description (cont'd)

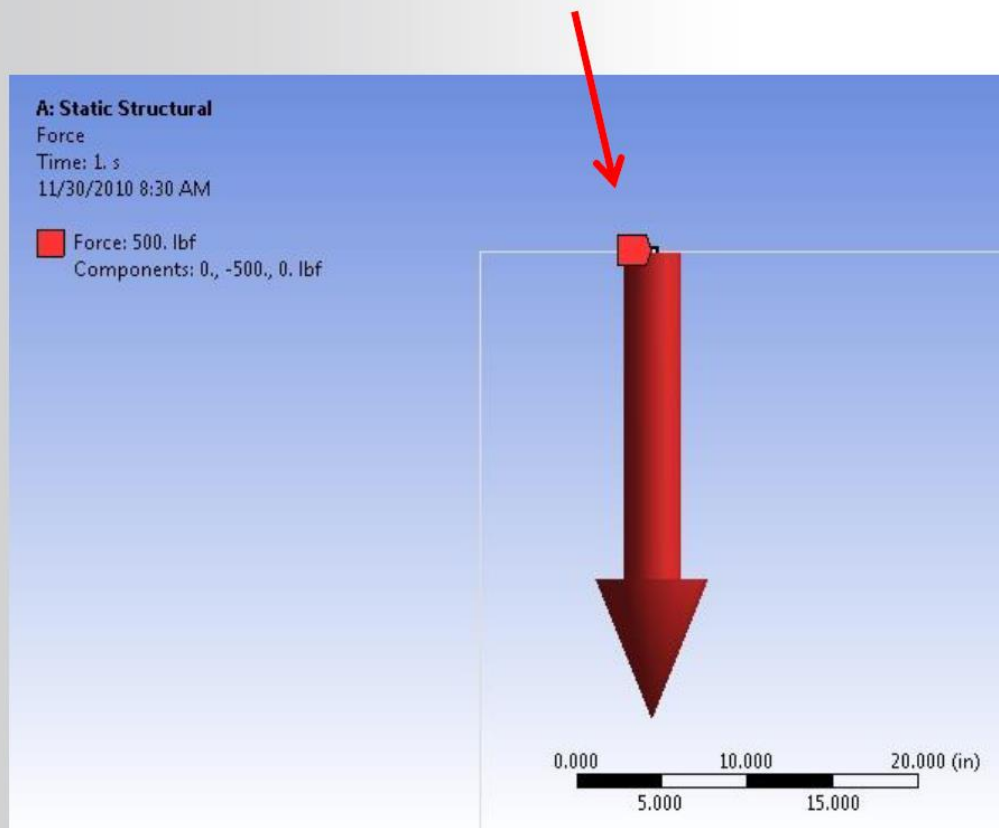
- Horizontal beam fixed at far end.
- Vertical column simply supported at bottom.
- All nodes are constrained from movement in Z direction.



...Workshop 5B: Post Buckling

- Model Description (cont'd)

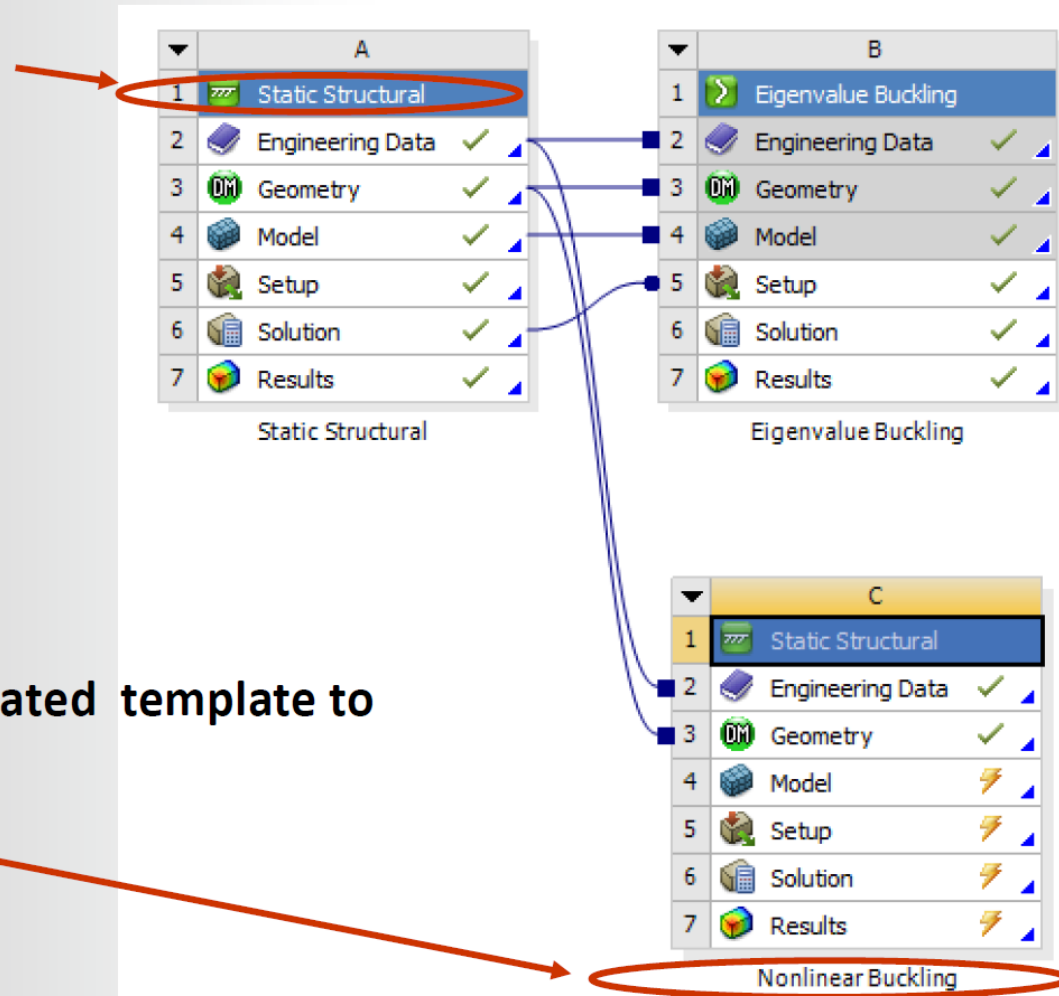
- 500 lb compressive load will be applied 10 inches off center of vertical column to induce buckling.



... Workshop 5B: Post Buckling

Steps to Follow:

After completing Workshop 5A, highlight the Model cell of the Static Structural Template and RMB “Duplicate” to produce the Project schematic below.

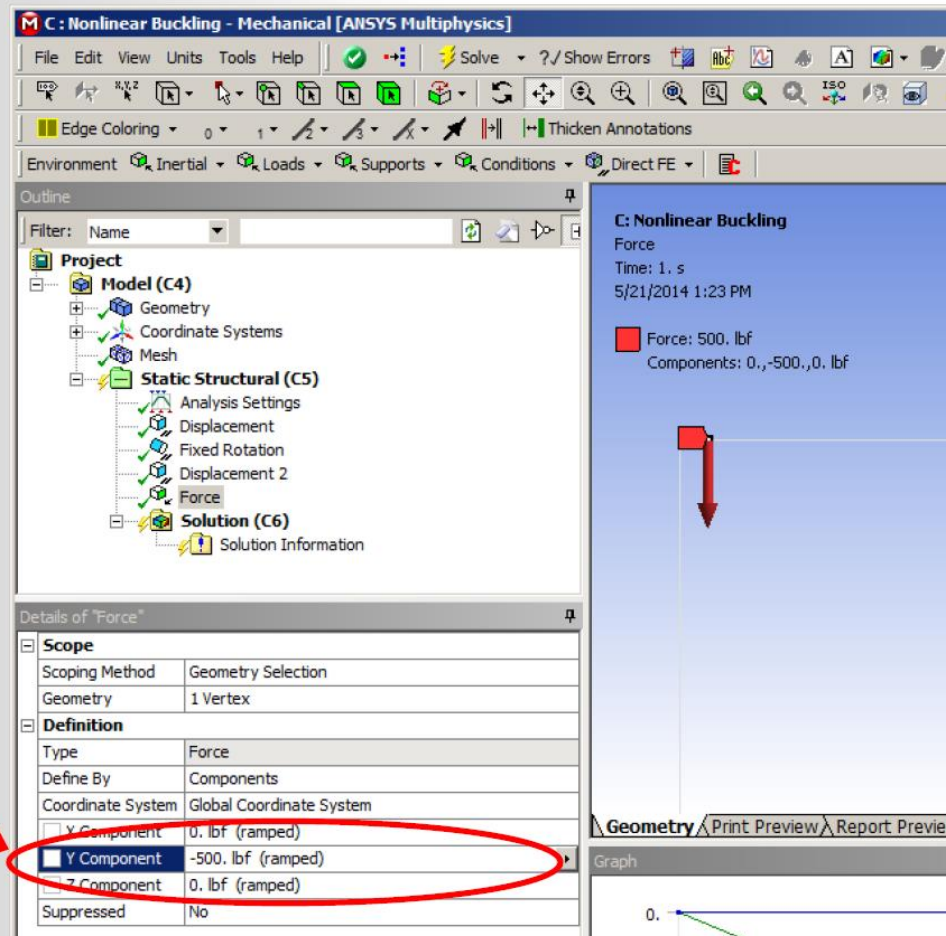


Change the Title of this newly created template to
“Nonlinear Buckling”

... Workshop 5B: Post Buckling

Open Mechanical Session for the Nonlinear Buckling.

- Highlight the Force Load and change the magnitude to 500lbf so as to induce a post buckling response.



Highlight Analysis Settings

Set (and/or confirm) the following specifications:

- Auto Time Stepping = ON
- Initial Substeps = 20 (min=20,max=1000)
- Weak Springs = OFF
- Large Deflection = ON,
- Generate Restart Points = Manual
- Load Step = All
- Substep = All
- Max Files to Save per Step = All
- Stabilization = OFF

Static Structural (C5)

Analysis Settings

Displacement

Fixed Rotation

Details of "Analysis Settings"

Step Controls	
Number Of Steps	1.
Current Step Number	1.
Step End Time	1. s
Auto Time Stepping	On
Define By	Substeps
Initial Substeps	20.
Minimum Substeps	20.
Maximum Substeps	1000.

Solver Controls	
Solver Type	Program Controlled
Weak Springs	Off
Large Deflection	On
Inertia Relief	Off

Restart Controls	
Generate Restart Points	Manual
Load Step	All
Substep	All
Maximum Points to Save Per Step	All
Retain Files After Full Solve	Yes

Nonlinear Controls	
Newton-Raphson Option	Program Controlled
Force Convergence	Program Controlled
Moment Convergence	Program Controlled
Displacement Convergence	Program Controlled
Rotation Convergence	Program Controlled
Line Search	Program Controlled
Stabilization	Off

Output Controls

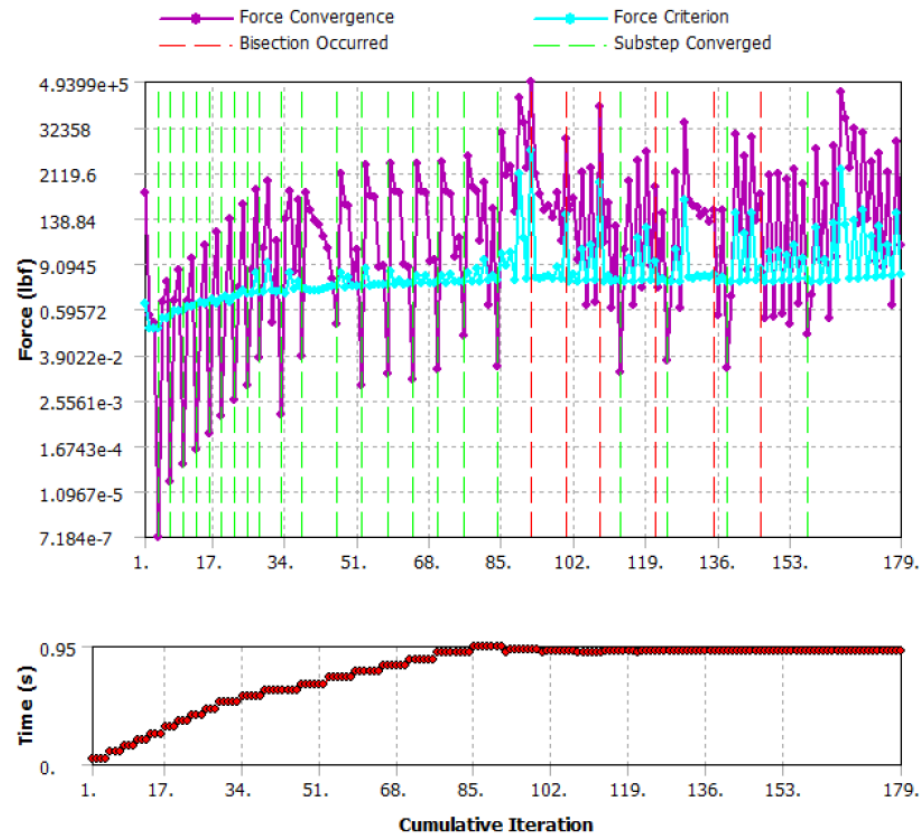
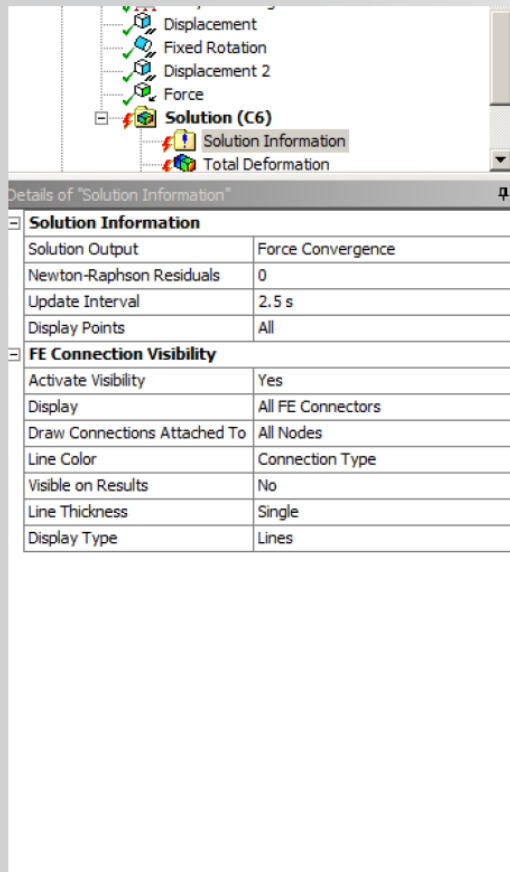
Analysis Data Management

... Workshop 5B: Post Buckling

Run the Solution



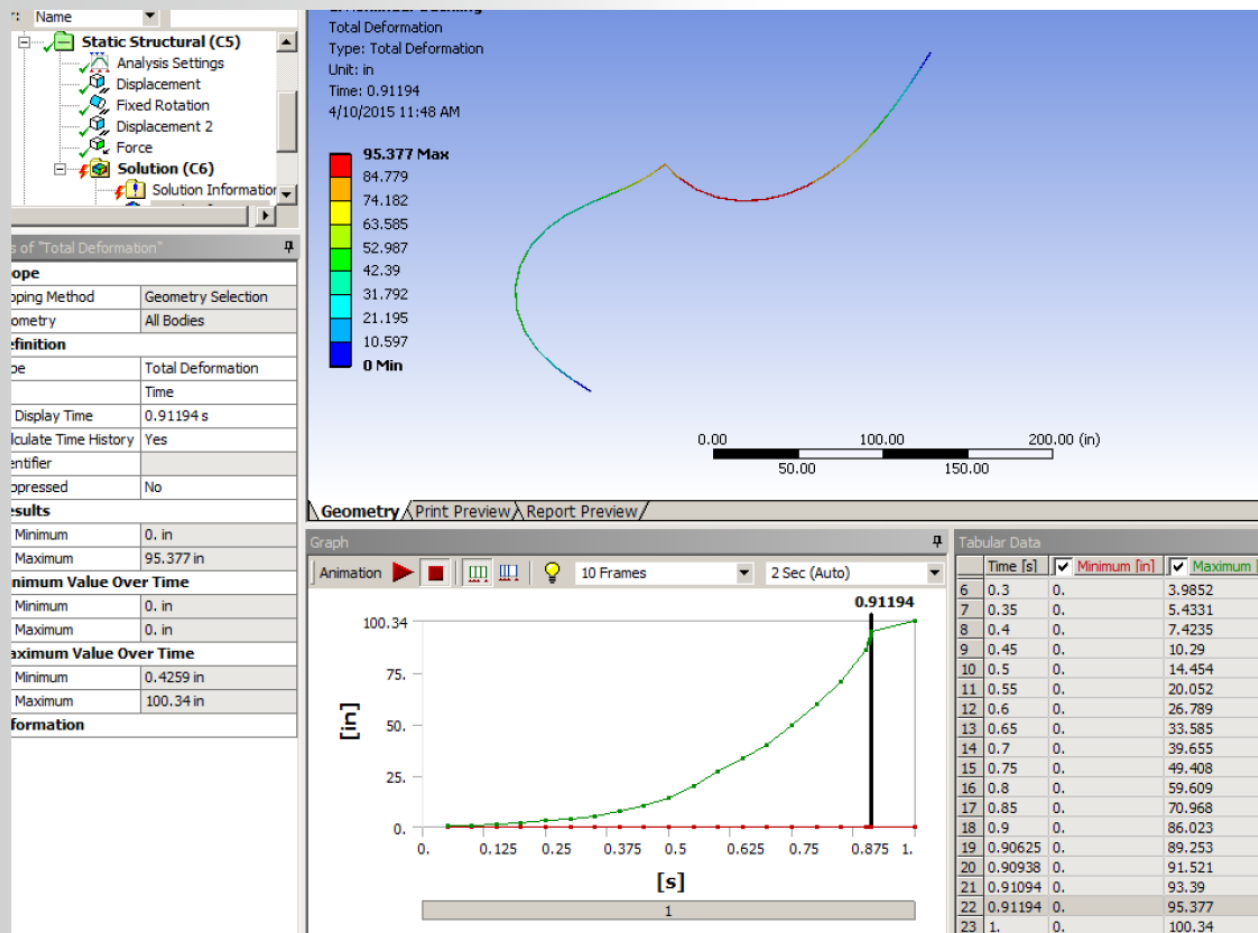
After a few bisections, the model fails to converge at less then 95% of the total load



... Workshop 5B: Post Buckling

Review the results from last successfully converged substep

- At the point of convergence failure, the structure is beginning to collapse onto itself and losing its stability



... Workshop 5B: Post Buckling

Highlight Analysis Settings and set up for a manual restart from last successfully converged solution utilizing Stabilization on non-convergence as specified

Restart specifications:

- Restart Type = Manual
- Restart Point = Load Step 1, Substep 22

Stabilization specifications:

- Constant
- Damping
- Damping Factor = 0.0001
- Activation ...= On Nonconvergence

Execute the restart...



Static Structural (C5)

- Analysis Settings
 - Displacement
 - Fixed Rotation
 - Displacement 2
 - Force
- Solution (C6)
 - Solution Information
 - Total Deformation

Details of "Analysis Settings"

Restart Analysis	
Restart Type	Manual
Action (Beta)	Continue Analysis
Current Restart Point	Load Step 1, Substep 22
Load Step	1
Substep	22
Time	0.91194 s

Step Controls

Solver Controls

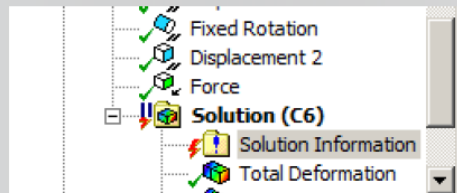
Restart Controls	
Generate Restart Points	Manual
Load Step	All
Substep	All
Maximum Points to Save Per Step	All
Retain Files After Full Solve	Yes

Nonlinear Controls

Newton-Raphson Option	Program Controlled
Force Convergence	Program Controlled
Moment Convergence	Program Controlled
Displacement Convergence	Program Controlled
Rotation Convergence	Program Controlled
Line Search	Program Controlled
Stabilization	Constant
--Method	Damping 1.1
--Damping Factor	1.e-004
Activation For First Substep	On Nonconvergence

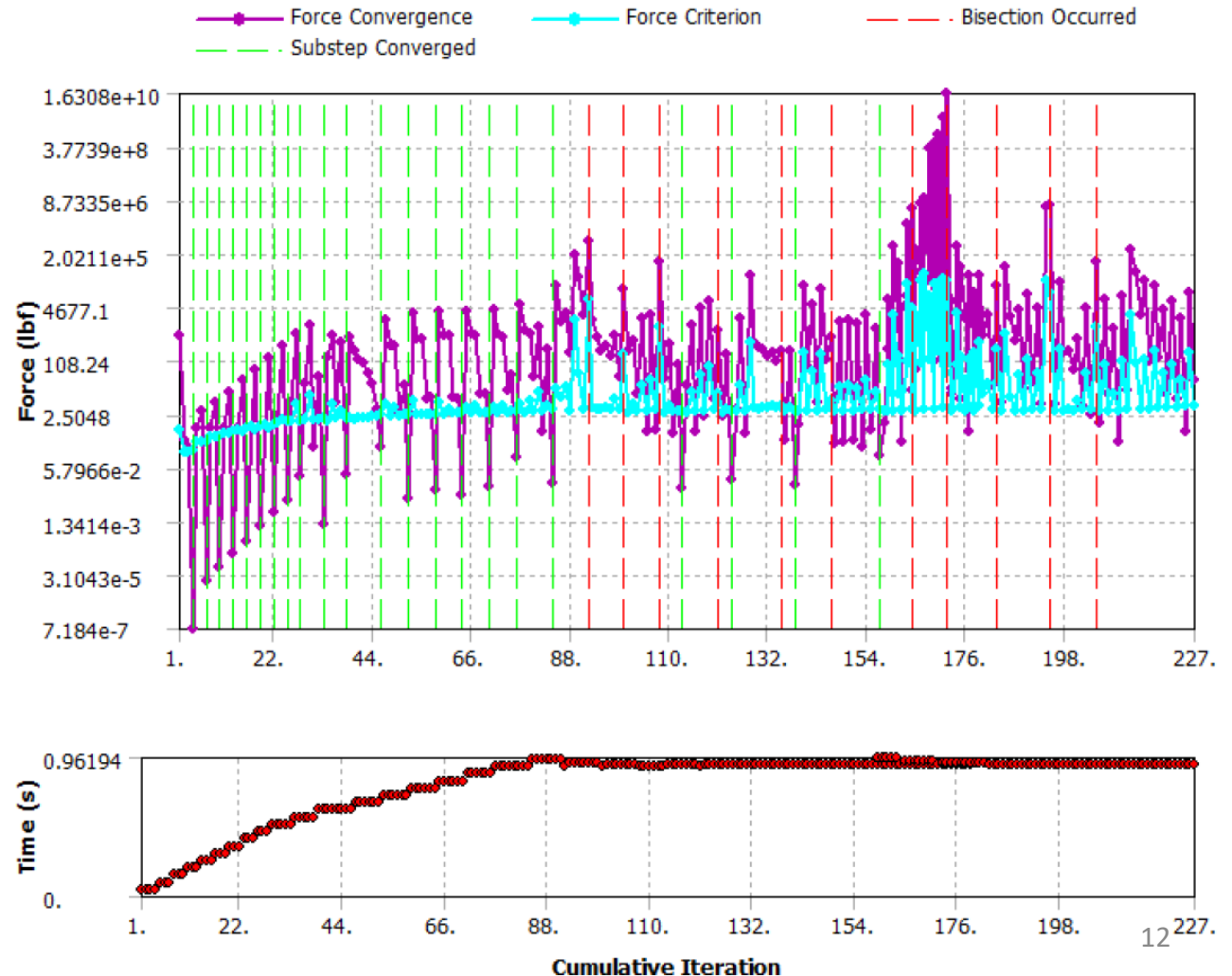
... Workshop 5B: Post Buckling

After a few bisections, the model still fails to converge



Details of "Solution Information"

Solution Information	
Solution Output	Force Conver...
Newton-Raphson Residuals	0
Update Interval	2.5 s
Display Points	All
FE Connection Visibility	
Activate Visibility	Yes
Display	All FE Connect...
Draw Connections Attached To	All Nodes
Line Color	Connection Type
Visible on Results	No
Line Thickness	Single
Display Type	Lines



... Workshop 5B: Post Buckling

Return to Analysis Settings and try to restart with stabilization from an earlier substep

Restart specifications:

- Restart Type = Manual
- *Restart Point = Load Step 1, Substep 20

Stabilization specifications:

- Constant
- Damping
- Damping Factor = 0.0001
- Activation ...= On Nonconvergence

Execute the restart...



* Stabilization is often a trial and error process. Had we studied the original solver output we would have found that convergence trouble began shortly after LS17 as evidenced with the beginning of many bisections.

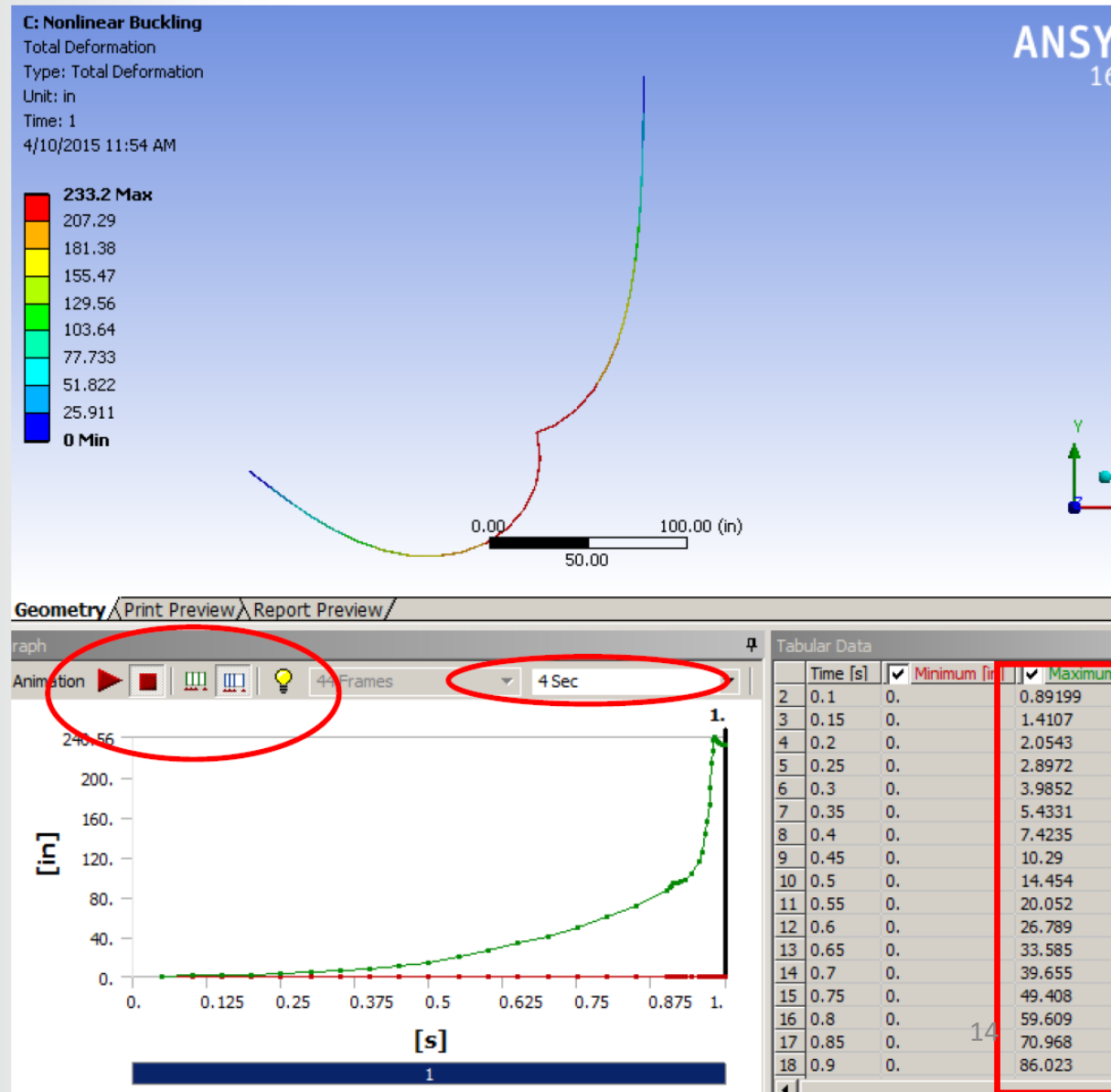
Static Structural (C5)

- Analysis Settings
- Displacement
- Fixed Rotation
- Displacement 2
- Force

Details of "Analysis Settings"

Restart Analysis	
Restart Type	Manual
Action (Beta)	Continue Analysis
Current Restart Point	Load Step 1, Substep 20
Load Step	1
Substep	20
Time	0.90938 s
Step Controls	
Solver Controls	
Restart Controls	
Generate Restart Points	Manual
Load Step	All
Substep	All
Maximum Points to Save Per Step	All
Retain Files After Full Solve	Yes
Nonlinear Controls	
Newton-Raphson Option	Program Controlled
Force Convergence	Program Controlled
Moment Convergence	Program Controlled
Displacement Convergence	Program Controlled
Rotation Convergence	Program Controlled
Line Search	Program Controlled
Stabilization	Constant
--Method	Damping
--Damping Factor	1.e-004
--Activation For First Substep	On Nonconvergence
--Stabilization Force Limit	0.2
Output Controls	
Analysis Data Management	

- Solution now converges
- Retrieve total displacement and other results at end time.
- Animate results
 - Over results set
 - Highlight the last 10% of displacement
 - Slow down the animation to 4 seconds per frame to get a better view of the snap thru behavior



... Workshop 5B: Post Buckling

- Use the Chart tool and user defined results to graph the nonlinear reaction force vs displacement in y direction at the point of load application

